

# Polyvibe

A lamp-and-photocell vibe: four unequal phase stages swept by a modelled light bulb, with a preamp for colour.



## Overview

Polyvibe is a vibe effect in the spirit of the classic lamp-and-photocell units. Four unequal phase-shift stages are swept by a light bulb shining on light-dependent resistors. The bulb heats up quickly and cools slowly, and the photocells lag behind it, so the sweep is lopsided - the familiar throbbing, liquid swirl rather than an even phaser.

In CHORUS mode the shifted signal is mixed with the dry one for the classic swirl; in VIBRATO mode you hear only the shifted signal, which gives a pitch wobble. A transistor-style preamp runs in every mode, adding asymmetric saturation, a low-mid lift and a rounded top - and PREAMP mode switches the vibe off so you can use the preamp on its own.

## Reach for it for

- The classic swirling vibe on clean or driven guitar
- Organ and electric piano that needs slow, throbbing movement
- Pitch-wobble vibrato on vocals, guitars and synths
- Wide stereo movement with SPREAD
- Warm preamp colour on its own (PREAMP mode)

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## Controls

### Modes (header)

|                |                                                                                                                        |
|----------------|------------------------------------------------------------------------------------------------------------------------|
| <b>CHORUS</b>  | The dry signal plus the phase-shifted one, half and half: the classic vibe swirl, with moving notches in the response. |
| <b>VIBRATO</b> | The phase-shifted signal only. The response stays flat and you hear a pitch wobble.                                    |
| <b>PREAMP</b>  | The vibe is off; only the preamp colours the sound. The lamp shows LAMP OFF.                                           |

### The display

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lamp</b>     | The glowing orb shows the bulb's brightness as it pulses, with L and R bars beside it for each channel (they differ when SPREAD is up). |
| <b>Response</b> | In CHORUS mode, the live frequency response with the current notch frequency shown above it. In VIBRATO mode the response is flat.      |

### Rate mode

|             |                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>FREE</b> | SPEED is set in Hz.                                                                                                         |
| <b>SYNC</b> | SPEED is a note value locked to the project tempo and timeline, so the sweep lines up with the bars and follows loop jumps. |
| <b>FLOW</b> | Tempo-synced, but the sweep runs continuously through loops and restarts.                                                   |

### Preamp

|               |                                                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AMOUNT</b> | The preamp colour, 0 - 100% (starts at 40%). It adds asymmetric saturation, lifts the low mids and rounds off the top. The word beside the knob shows the character: clean, |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

warm, pushed or driven. At 0% the preamp is clean. Works in every mode.

## Vibe

|                  |                                                                                                                                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SPEED</b>     | How fast the lamp pulses: 0.1 - 12 Hz in FREE, or a note value from 4 BARS to 1/32 (with dotted and triplet values) in SYNC and FLOW.                                                                |
| <b>INTENSITY</b> | How far the sweep travels, 0 - 100%. Low settings are subtle; high settings are deep and swirling.                                                                                                   |
| <b>THROB</b>     | How much of the bulb's lopsided character you hear, 0 - 100%. At 0% the sweep follows a smooth sine; turned up, the fast-heat, slow-cool lamp and lagging photocells give the uneven, pulsing throb. |
| <b>SPREAD</b>    | Offsets the right channel's sweep from the left, 0 - 180 degrees. 0 is mono movement; higher settings make the vibe move across the stereo field.                                                    |

## Drift

A slow random wander, like an old lamp and tired components. DRIFT (0 - 100%, starting at 50%) sets how much: 50% is the standard amount, 100% twice that, 0% none. In Polyvibe it moves three things: the speed (in FREE the rate wanders; in SYNC and FLOW the sweep wobbles but stays locked on average), the intensity (the sweep gets a little shallower at times) and the lamp brightness (the bulb dims a little). The wander also speeds up a little as DRIFT goes up.

## IN / OUT

|              |                                                                                                                                                                                                                                                                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TRIM</b>  | Input level before anything else, from -24 to +12 dB.                                                                                                                                                                                                                                                                                                         |
| <b>GUARD</b> | Click the TRIM knob (without moving it) to switch the guard; its light sits at the knob's upper right. On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The light glows brighter while it is catching peaks. |

|                                 |                                                                                                                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SAT</b>                      | Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.                                                                         |
| <b>OFF / CLIP / LIMIT (OUT)</b> | The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). CLIP is on by default; its key flashes red while it is clipping. |

## Master

|               |                                                                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MIX</b>    | Blends the untreated input back in, 0 - 100%. At 100% you hear only the effect (which, in CHORUS mode, already contains the dry signal). |
| <b>OUTPUT</b> | Output level, -24 to +12 dB.                                                                                                             |
| <b>Meters</b> | Output level, left and right.                                                                                                            |

## The header

|                  |                                                                                                                                                                                                                                                                                                            |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PRESETS</b>   | Opens the list of factory presets; the ◀ / ▶ keys step through them and the display shows the one in use. Choosing one sets the mode, rate, vibe, preamp, MIX and drift settings; the DRIFT dial, OUTPUT and the IN / OUT controls are left as they are. The last entry resets everything to the defaults. |
| <b>S / M / L</b> | Panel size: small, medium or large. The panel also scales to fit the plugin window.                                                                                                                                                                                                                        |
| <b>?</b>         | Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.                                                                                                                                                                                 |

## Presets

|                       |                                                                   |
|-----------------------|-------------------------------------------------------------------|
| <b>Classic Chorus</b> | The everyday vibe swirl: CHORUS at 2 Hz, 70% intensity and throb. |
|-----------------------|-------------------------------------------------------------------|

|                              |                                                                                                   |
|------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Slow Throb</b>            | A slow, deep CHORUS (0.6 Hz) with strong throb and a little more lamp drift.                      |
| <b>Fast Swirl</b>            | A quick CHORUS at 6.5 Hz, a little less deep.                                                     |
| <b>Vibrato</b>               | VIBRATO mode at 4.5 Hz for a pitch wobble, with a lighter preamp.                                 |
| <b>Stereo Vibe</b>           | CHORUS at 1.5 Hz with the right side offset by 90 degrees.                                        |
| <b>Synced 1/8</b>            | CHORUS locked to eighth notes of the project tempo.                                               |
| <b>Preamp Only (Clean)</b>   | The vibe off, with a light preamp colour.                                                         |
| <b>Preamp Only (Pushed)</b>  | The vibe off, with the preamp at 80%.                                                             |
| <b>Ageing Lamp</b>           | A slow, throbbing CHORUS with lots of drift on the speed, depth and lamp - an old, unsteady unit. |
| <b>Reset all to defaults</b> | Every control back to its starting value.                                                         |

## Tips

- For the classic guitar vibe, start with Classic Chorus and set SPEED by ear to the tempo of the song; turn THROB up for more of the pulsing character.
- Put Polyvibe before a drive or amp for a thick, chewy swirl; after it for a cleaner, more open sound.
- VIBRATO with a low INTENSITY gives a gentle, tape-like wobble on vocals and keys.
- On a mono source, a SPREAD of 90 degrees or more makes the movement swirl across the speakers.

File: gfx-polyvibe.jsfx